DELIVERABLE D3.3

Use Case Scenarios and KPIs Report

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SEMDR

Smart Energy Management and Demand Response

ANDREY SESYUK, KARIM ARNOUS

IMKA HOLDINGS LTD

ANDREAS PROCOPIOU

GRIDQ LTD

GIORGOS MASOUREKKOS

RESTROL

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Abstract

This report constitutes Deliverable D3.3 of the SEMDR project (Smart Energy Management & Demand Response Capabilities for Sustainable Industry), produced under Work Package 3, Task T3.3. It defines the use case scenarios and Key Performance Indicator (KPI) framework that will govern the validation of the SEMDR system at the pilot site in Limassol, Cyprus.

The document establishes four core use cases: (i) pool-pump duty-cycle optimisation, (ii) HVAC supervisory set-point tuning in shared spaces, (iii) refrigeration efficiency investigation, and (iv) peak-period demand shaping; each grounded in the real operational context of the pilot site as observed and instrumented during Months 1-6. For each use case, the report specifies the trigger conditions, controllable loads, safeguards protecting guest comfort and food safety, and the primary KPIs used for assessment.

The KPI framework is structured in three layers: Technical KPIs that confirm platform health and data integrity, Energy and Cost KPIs that quantify consumption and bill impact, and Service and Comfort KPIs that protect the hotel's operational quality. Every KPI is defined with its measurement formula, sampling window, and acceptance threshold, enabling a direct and auditable comparison of results in Deliverable D3.4.

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1 Introduction

1.1 Purpose and Scope

Deliverable D3.3 defines testable use case scenarios and a quantitative KPI framework for evaluating the performance of the SEMDR energy management system at the pilot site. The report serves as the formal measurement specification for the project. It translates the system requirements established in D3.2 and the state-of-the-art findings in D3.1 into a concrete validation plan: what will be tested, how it will be measured, and what constitutes success. Every KPI defined in this document will be directly referenced in D3.4 (System Validation and Performance Report), enabling a rigorous like-for-like comparison between targets and achieved outcomes. Table 1 defines the relationship to all other deliverable parts of the project:

Deliverable	Title	Relationship to D3.3
D3.1	State of the Art Review (M2)	Provides the evidence base for the control strategies and comfort benchmarks adopted in the use cases
D3.2	Technical Requirements & Specifications (M3)	Defines the system architecture, integration points, and non-functional targets that constrain use case design
D1.3	Interim Progress Report (M6)	Reports actual operational data (baselines, telemetry, trial results) that inform the KPI thresholds in this document
D4.1	Data Collection & Analysis Report (M4)	Provides asset inventory and data-path description; confirms instrumented systems available for KPI measurement
D4.2	Predictive Algorithm Development Report (M8)	Shadow-mode model outputs will be compared against D3.3 KPI thresholds before promotion to actuation
D3.4	System Validation & Performance Report (M12)	Primary consumer of this document - D3.3 defines the framework against which D3.4 reports actual results
D5.2	System Validation Report (M11)	WP5 field validation will use the same KPI framework for the integrated system performance assessment

Table 1 - D3.3 Relation to Other Deliverables

The scope covers four use cases selected on the basis of energy impact, safety of operation, and technical readiness at the pilot site. The document defines scenarios, triggers, safeguards, and KPI formulas for each, and specifies a staged validation methodology aligned with the project's principle of gated progression from shadow operation to controlled actuation.

Section 3 describes the pilot site context and the operational baseline established through live telemetry. Section 4 defines the four use case scenarios. Section 5 presents the multi-layer KPI framework with formulas, sampling windows, and thresholds. Section 6 specifies the validation methodology including the staged activation protocol and acceptance criteria. Section 7 concludes with the link to D3.4.

2 System Context and Pilot Site Overview

2.1 Pilot Site: Elias Beach Hotel

The SEMDR system is being deployed at Elias Beach Hotel, a four-star resort located in Limassol, Cyprus. The hotel is a full-service hospitality property operating year-round with seasonal variation in occupancy and energy demand driven by the Mediterranean climate. It was selected as the pilot site on the basis of a signed Letter of Intent and confirmed technical access to key energy subsystems.

The hotel's energy profile is characteristic of Cyprus resort properties: high base loads sustained by HVAC, refrigeration, and water systems, overlaid with occupancy-driven peaks during daytime and evening hours. Electricity is procured from the grid under a tariff structure that includes time-of-use and peak demand components, creating clear economic incentives for demand response. The site presents diverse controllable assets - pool circulation pumps, chiller-based HVAC systems, commercial refrigeration circuits, and mains-level aggregated demand - that collectively offer material flexibility potential.

Access protocols and safety procedures have been agreed with hotel operations management. All on-site work is conducted during evening or other low-impact windows to ensure zero disruption to guests. Hotel staff have been briefed on the system and its operating principles.

Figure 1 - Elias Beach Hotel

2.2 Monitored Assets and Integration Points

Table 2 showcases the instumented and monitored assets as part of the SEMDR deployment. Integration methods and data sources are noted for each.

Asset / System	Integration Method	Data Streams	Status (at M6)
Hotel mains supply	SELEC MFM376 three-phase meter integrated at existing compliant main board; Raspberry Pi edge gateway via Modbus/RS-485; data transmitted at 1-minute intervals via MQTT over TLS 1.2/1.3 to AWS IoT Core	Total active power (kW), daily energy (kWh), phase voltages, currents	Live - continuous since August 2025 (M5)
Pool circulation pumps (860 m^3 pool, 12 kW, 80 m^3/h, 3.6h turnover)	SELEC MFM376 on pump board; Raspberry Pi with relay board for remote ON/OFF control; EcO Start + Connect pool quality sensor in overflow tank (pH, ORP, temperature)	Pump active power (kW), energy (kWh); pool pH, ORP, temperature (1-minute intervals)	Live - continuous since M5 (August 2025); supervised duty-cycle control trials executed M6 (September 2025)
HVAC - chiller BMS supervisory layer	BMS oBIX API integration for real-time data reading and remote set-point adjustment; edge gateway BMS interface; total installed chiller capacity 720 kW	Supervisory set-points, zone temperatures, chiller run/status signals via oBIX API (real-time)	BMS upgrade completed M6; metering installation in progress; power telemetry expected upon completion.
Refrigeration circuits	SELEC MFM376 three-phase meter paired with temperature controller at refrigeration board; monitoring only	Compressor power, temperature set-point and measured value (1-minute intervals)	Live - continuous metering active; under investigation as DR target (see Section 3.3)
Lobby environment	Shelly H&T Gen 3 temperature and humidity sensor (main lobby)	Dry bulb temperature (°C), relative humidity (%) - continuous	Live - continuous since M5 (August 2025)
Weather (external)	AccuWeather API integration; hourly forecast and actuals ingested into feature pipeline	Ambient temperature, humidity, solar irradiance forecast, wind speed/direction	Live - integrated in feature pipeline since M5

Table 2 - Monitored Assets and Integration Points

2.3 Operational Data and Baseline

Continuous telemetry has been live since August 2025 for the mains, pool pump, and refrigeration boards. The following baseline observations, drawn from the data reported in D1.3, define the reference conditions against which intervention outcomes will be measured.

Main Power Baseline
The main supply meter records a daily operating band of approximately 135 kW (night-time and early morning minimum) to 180 kW (afternoon and evening peak). A mid-month trough near 135-140 kW has been observed, consistent with slightly lower occupancy, and isolated peaks just below 180 kW are characteristic of concurrent HVAC and kitchen load. This profile establishes the baseline against which aggregate demand reduction will be measured under UC-4.

Figure 2 - Pool Pump Baseline Load Profile

Pool Pump Baseline
The pool circulation pump (12 kW three-phase, 80 m^3/h flow rate, 3.6-hour full-pool turnover time, serving an 860 m^3 outdoor pool) presents a steady baseload under normal 24/7 operation. At this operating point the pump draws approximately 288 kWh per day, representing close to 10% of instantaneous hotel demand when mains power holds in the 135-150 kW range. This steady, predictable profile makes the duty-cycle use case both measurable and low-risk. Conservative hand-calculated estimates: runtime reductions of 30 minutes per day yield approximately 6 kWh/day (~180 kWh/month); 1 hour yields ~12 kWh/day (~360 kWh/month); 2 hours yields ~24 kWh/day (~720 kWh/month). These conservative figures serve as the formal baseline for KPI target-setting purposes; algorithmic optimisation using tariff-aware scheduling is expected to improve on these estimates as models mature in the second project period.

Figure 3 - Pool Pump Baseline Load Profile

Water Quality Baseline
Pool water quality is monitored continuously via EcO Start + Connect, measuring pH, ORP (oxidation-reduction potential), and temperature in the pool overflow tank. During the supervised pump duty-cycle trials conducted in Month 6 (September 2025), electrical telemetry was combined with water-quality readings to confirm that runtime reductions at durations of 30 minutes, 1 hour, 2 hours, 3 hours, and 4 hours did not materially degrade ORP or cause pH to breach operational bounds. The safety floor ORP threshold is set at 650 mV [1, 2]. The algorithm design also incorporates a chemical dosing detection mode: on days when chlorine treatment is being administered, the system will enter 'dispersion mode' to ensure adequate chemical distribution before any pump OFF periods are scheduled. This provides the evidence basis for the water-quality safeguard thresholds in UC-1.

Lobby Comfort Baseline
The Shelly H&T Gen 3 sensor installed in the main lobby records dry bulb temperature and relative humidity. Initial data confirm that the lobby operates within the target comfort band of 22-24°C during service hours. This provides the reference for the guest comfort KPI in UC-2.

3 Use Case Scenarios

Four use cases have been defined for the validation with each being grounded in the asset inventory and operational constraints of the pilot site. Use cases were selected on the following criteria: (a) material energy impact relative to overall hotel consumption; (b) availability of direct measurement before and after intervention; (c) ability to implement reversible, bounded control without risk to guests or operations; and (d) readiness of the underlying telemetry and control infrastructure by the time validation commences. Each use case is presented below as a structured card summarising the key scenario parameters, followed by a detailed narrative. The KPIs associated with each use case are fully defined in Section 4.

3.1 UC-1: Pool-Pump Duty Cycle Optimisation

UC-1: Pool-Pump Duty-Cycle Optimisation	
Summary	Reduce pool pump runtime during low-impact periods using AI-generated duty-cycle recommendations, while maintaining pool water quality within safe operational bounds.
Trigger / Activation	Short-horizon demand forecast anticipates a period where load reduction is beneficial (e.g., approach to daily peak), or scheduled daily optimisation window is reached.
Controllable Loads	Pool circulation pump (approximately 12 kW, three-phase). Relay board enables remote on/off control. No guest-facing HVAC or climate systems are involved.
Safeguards	Automatic reversion to continuous operation if pool ORP drops below 650 mV or pH moves outside 7.2-7.8 [1]. Maximum cumulative off-time capped at 4 hours continuous (validated safe in supervised trials). On days when chemical dosing is detected, the algorithm will enter dispersion mode - pump must run continuously for a sufficient window before any OFF period is scheduled. Emergency stop available at hotel panel and via dashboard at all times. Water quality readings must be confirmed live and within bounds before any off-period begins.
Primary KPIs	Daily pump energy (kWh); daily demand reduction from pump (kW); tariff-period energy savings (€); ORP maintenance ≥ 650 mV; pH 7.2-7.8; pump runtime utilisation (%)

This use case represents the most mature and best-evidenced intervention in the project portfolio. The pool pump was the first load to be fully instrumented, with both electrical metering and water-quality monitoring operational from Month 5 (August 2025). Supervised trials executed during Month 6 (September 2025) - involving daily remote on/off cycles at durations of 30 minutes, 1 hour, 2 hours, 3 hours, and 4 hours across a one-week window - confirmed that the control path, logging, and water-quality safeguards all behaved as designed. The trials demonstrated that runtime reductions up to 4 hours produce no measurable degradation in ORP or pH when rescheduling windows are tuned to low-impact periods.

The optimisation approach will use a duty-cycle scheduling algorithm that combines water-quality modelling with tariff-aware cost minimisation [2]. The algorithm design incorporates a multi-state ORP model to predict water quality evolution under different pump runtime scenarios, enabling the system to identify the maximum safe off-time for each scheduling

window. Price signals will be sourced from TSOC Cyprus day-ahead market clearing prices when available, with the current EAC tariff structure used in the interim. Feature inputs will include current pump power draw, pool pH, ORP and temperature, lobby ambient conditions, external weather data, and tariff signals. The algorithm will execute on rolling scheduling windows via the Celery task scheduler integrated into the Django platform stack (as specified in D3.2). Recommendations will initially be generated in shadow mode for operator review; progression to autonomous execution within defined guardrails will follow the staged validation protocol defined in Section 5.

The conservative hand-calculated savings estimates from the M6 baseline (Section 2.3) represent the minimum expected benefit from simple runtime reduction. Tariff-aware scheduling - concentrating off-periods during high-price windows - is expected to deliver additional cost savings beyond the volumetric energy reduction. The magnitude of this tariff benefit will depend on the price differential available in the TSOC day-ahead market and will be quantified during the validation period.

3.2 UC-2: HVAC Supervisory Set-Point Tuning in Shared Spaces

UC-2: HVAC Supervisory Set-Point Tuning in Shared Spaces	
Summary	Adjust supervisory HVAC set-points and schedules in shared hotel areas within comfort bands, prioritising changes during off-peak tariff periods and pre-cooling the space ahead of peak demand periods.
Trigger / Activation	Tariff-period transition (peak tariff window approaching within 30-60 minutes), scheduled pre-cooling window, or model-identified opportunity to reduce chiller load during a low-occupancy period. Activation only within defined comfort bounds and operating hours.
Controllable Loads	Chiller BMS supervisory set-point for shared zones. Changes are bounded to ±1°C from the nominal set-point.
Safeguards	Lobby temperature must remain within 22-24°C throughout any intervention. Set-point change limited to ±1°C from nominal. Automatic reversion if measured temperature drifts outside the comfort band. BMS monitors all changes and can override immediately. Changes are logged with version, timestamp, and measured outcome.
Primary KPIs	Chiller energy reduction (kWh) vs. baseline; peak-period chiller power (kW); lobby temperature compliance (%); set-point deviation from nominal (°C); pre-cooling effectiveness (°C·h saved during peak tariff window)

This use case targets the hotel's chiller system, which at 720 kW total installed capacity is the single largest controllable load on the site. Access to the chiller BMS required a vendor-approved upgrade, which was secured late in Month 6 (September 2025). With the upgrade complete, consumption meters are being installed on the rooftop chiller feeds at the time of this document. Production-quality chiller power telemetry is expected once metering installation is complete. The BMS interface will enable both real-time reading of operational data (supervisory set-points, zone temperatures, chiller run/status signals) and remote adjustment of supervisory temperature set-points for shared zones.

The control strategy will focus on two sub-scenarios. The first is pre-cooling: advancing the cooling schedule by 1-2 hours before anticipated peak tariff periods, storing thermal energy in the building fabric and reducing compressor demand during the peak window. The second is conservative set-point relaxation during low-occupancy periods (e.g., early morning before guest activity), allowing the chiller to rest while the lobby remains within the upper comfort bound. Both sub-scenarios are reversible and will operate exclusively within the ±1°C guard on the nominal set-point [3].

Given that chiller power telemetry will stabilise from Month 7 (October 2025), the energy consumption baseline for UC-2 will be established using the first 4-6 weeks of production-quality chiller metering data. This will be clearly noted in D3.4, with the energy KPI comparison window adjusted accordingly. The lobby comfort sensor and the chiller supervisory data provide continuous independent verification of thermal conditions and set-point adherence throughout all interventions.

3.3 UC-3: Refrigeration Efficiency Investigation

UC-3: Commercial Refrigeration - Under Investigation	
Summary	Continuous monitoring of commercial refrigeration circuits is in place. The consortium is assessing the feasibility of refrigeration as a demand response target based on preliminary data collected during Months 5-6. Findings and a final determination on DR viability will be reported in D3.4 (Month 12).
Trigger / Activation	N/A for demand response actuation at this stage. Monitoring continues to detect abnormal compressor cycling or temperature deviations that may surface advisory recommendations to hotel operations.
Controllable Loads	Commercial refrigeration compressor circuits monitored via SELEC MFM376 and temperature controller. No actuation - read-only monitoring only.
Safeguards	No direct compressor actuation under any circumstances at this stage. Temperature readings are logged continuously. Any deviation from safe food-storage temperature bands triggers an immediate alert to hotel operations staff. Food safety is the overriding constraint and is never compromised.
Primary KPIs	To be confirmed following investigation - preliminary indicators under consideration include compressor duty-cycle efficiency, temperature stability within HACCP bounds, and anomaly detection accuracy. Final KPI definitions will be reported in D3.4.

Continuous monitoring of commercial refrigeration circuits has been active since Month 5 (August 2025). Preliminary observations indicate three factors that may limit refrigeration's viability as a meaningful DR asset within the project scope:

- Tight food-safety temperature bands (1-2°C permitted deviation) leave limited room for load shifting or set-point adjustment without creating risk of HACCP non-compliance [1].
- Commercial refrigeration units operate on a near-continuous duty cycle, making duty-cycle-type optimisation (as applied to the pool pump) potentially inappropriate - the compressor may not tolerate meaningful off-periods.

- Individual refrigeration circuit power is comparatively low relative to the pool pump (12 kW) and the chiller plant (720 kW total), potentially limiting the energy impact of any intervention even if technically feasible.

These observations are preliminary. The consortium will continue monitoring and analysis through the second project period to reach a definitive assessment. If refrigeration proves unsuitable as a DR target, the monitoring data will still contribute value through anomaly detection - identifying inefficient cycling patterns or temperature deviations that indicate maintenance requirements.

3.4 UC-4: Peak-Period Demand Shaping

UC-4: Peak-Period Demand Shaping	
Summary	Combine short-horizon demand forecasts with tariff-period awareness to reduce the hotel's peak demand level and peak-tariff energy consumption by coordinating modest, reversible load adjustments across available controllable assets before and during peak windows.
Trigger / Activation	Short-horizon demand forecast projects that hotel mains demand will exceed a rolling 7-day 90th percentile within the next 30-60 minutes, or a high-tariff period is approaching. Demand shaping actions are composed from available UC-1 and UC-2 levers.
Controllable Loads	Combination of pool pump duty-cycle (UC-1) and HVAC pre-cooling / set-point relaxation (UC-2) coordinated as a portfolio. The system selects the lowest-disruption combination that achieves the targeted demand reduction.
Safeguards	All individual use-case safeguards apply without exception. Aggregate demand reduction limited to 15% of prevailing load to avoid uncomfortable guest-visible changes. Minimum inter-action interval of 30 minutes to avoid rapid cycling. Hotel operations can suspend demand shaping at any time via a single dashboard toggle. Automatic reversion if any comfort or safety KPI is breached.
Primary KPIs	Peak demand reduction (kW and %); peak-tariff energy reduction (kWh and €); peak-to-average ratio improvement; demand reduction activation success rate (%); guest comfort compliance throughout demand events (%)

This use case represents the highest-level integration of the SEMDR system, treating the hotel's energy portfolio as a coordinated flexibility resource [3]. It builds directly on the infrastructure and trust established through UC-1 and UC-2 and is expected to be the primary source of tariff-driven savings reported in D3.4.

The demand shaping logic will use the UC-1 pump scheduling algorithm augmented with a tariff-awareness layer drawing on TSOC Cyprus day-ahead market clearing prices, combined with the HVAC set-point control capability established in UC-2. The policy engine will select from the available UC-1 and UC-2 actions, subject to their individual guardrails, and compose a coordinated response. The portfolio approach is important because neither pump control nor HVAC tuning alone is likely to be sufficient to achieve the headline demand reduction target; their combination provides the required flexibility depth.

4 KPI Framework

4.1 Framework Architecture

The KPI framework is structured in three layers, reflecting the three dimensions along which the SEMDR system must deliver and be assessed: platform reliability, economic impact, and service quality. This layered structure ensures that high-level energy savings cannot mask underlying technical problems, and that cost reductions are not achieved at the expense of guest comfort.

Layer	Focus	Question Answered	Consequence if Failed
Technical KPIs	Platform health, data quality, control reliability	Is the system working correctly?	Invalidates energy and comfort results; requires technical remediation before continuing
Energy and Cost KPIs	Consumption reduction, demand management, bill impact	Is the system delivering economic value?	Project objectives not met; root cause analysis required
Service and Comfort KPIs	Guest temperature, humidity, food safety, operational workload	Is the system preserving service quality?	Intervention must be suspended immediately; safeguards reviewed

Table 3 - Three Layer Framework Architecture

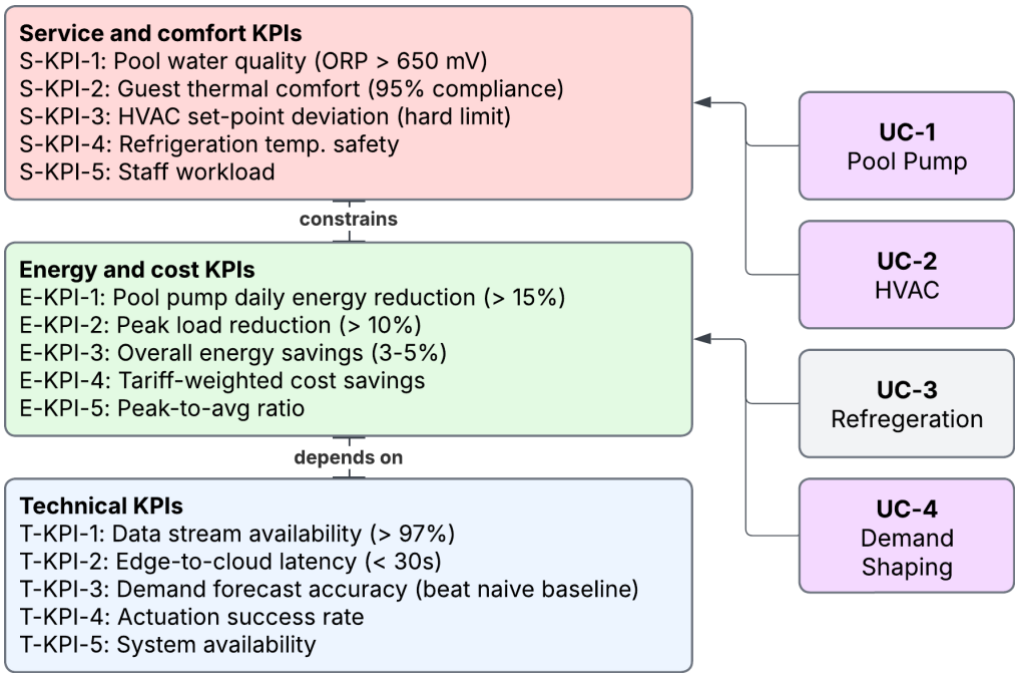

Figure 4 - SEMDR KPI Framework

A KPI passes only when it meets its acceptance threshold consistently across the full validation period, not just in isolated measurements. Each KPI is measured over a defined sampling window. For daily KPIs, the acceptance threshold must be met on at least 80% of operational days in the validation period to be considered achieved. For event-level KPIs, the threshold applies per event. Where a baseline is not yet available (notably for chiller data), this is explicitly noted and the KPI assessment will begin from the first full month of chiller telemetry.

4.2 Technical KPIs

4.2.1 T-KPI-1: Data Stream Availability

Measures the proportion of expected data points that are received and pass quality checks within the defined sampling window. This KPI confirms that the edge-to-cloud pipeline is operating reliably and that the data foundation for algorithmic decisions is sound.

T-KPI-1: Data Stream Availability	
Formula	$$\boldsymbol{Data\ Stream\ Availability}\ (\%) = \frac{\boldsymbol{N_{valid,\ received}}}{\boldsymbol{N_{expected}}} \times \mathbf{100}$$ $N_{\text{valid, received}}$ = number of valid samples received N_{expected} = number of expected samples
Sampling Window	Rolling 24-hour and rolling 7-day
Acceptance Threshold	≥ 97% over any rolling 7-day window for each instrumented asset
Applicable use cases	All (UC-1, UC-2, UC-3, UC-4)

4.2.2 T-KPI-2: Edge-to-Cloud Latency

Measures the time from data acquisition at the edge device to availability in the cloud time-series store. Excessive latency degrades forecast accuracy and can cause the control policy to act on stale information.

T-KPI-2: Edge-to-Cloud Latency	
Formula	$$\boldsymbol{Latency_{95}}\ [s] = \boldsymbol{P_{95}(t_{cloud_arrival} - t_{edge_acquisition})}$$ $t_{\text{cloud_arrival}}$ = cloud arrival timestamp $t_{\text{edge_acquisition}}$ = edge acquisition timestamp $P_{95}(\cdot)$ = 95th percentile
Sampling Window	Rolling 1-hour
Acceptance Threshold	95th percentile ≤ 10 seconds under normal operating conditions
Applicable use cases	UC-1, UC-2, UC-4 (latency-sensitive control); UC-3 (advisory, less critical)

4.2.3 T-KPI-3: Demand Forecast Accuracy (Short-Horizon)

Measures the accuracy of the short-horizon (15-minute ahead) demand forecast model. This KPI confirms that the forecaster is sufficiently accurate to support meaningful demand shaping decisions under UC-4.

T-KPI-3: Demand Forecast Accuracy (Short-Horizon)	
Formula	$$\boldsymbol{MAPE}\ (\%) = \frac{\mathbf{1}}{\boldsymbol{N}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \left\lvert \frac{\boldsymbol{y_i - \hat{y}_i}}{\boldsymbol{y_i}} \right\rvert \times \mathbf{100}$$ y_i = actual demand $\hat{y}_i$ = forecast demand N = number of forecast points in the evaluation window
Sampling Window	Rolling 7-day on 15-minute forecasts
Acceptance Threshold	MAPE ≤ 8% on rolling 7-day window; better than a naive persistence baseline
Applicable use cases	UC-4 (primary); UC-1 (secondary - pump scheduling)

4.2.4 T-KPI-4: Actuation Success Rate

Measures the proportion of authorised control commands that are executed successfully and confirmed by telemetry within the expected response window. Failed actuations may indicate relay faults, communication issues, or BMS integration problems.

T-KPI-4: Actuation Success Rate	
Formula	$\boldsymbol{Actuation\ Success\ Rate}\ (\%) = \frac{\boldsymbol{N_{successful,\ confirmed}}}{\boldsymbol{N_{authorised,\ issued}}} \times \mathbf{100}$ $N_{successful,\ confirmed}$ = number of authorised commands successfully executed and confirmed by telemetry $N_{authorised,\ issued}$ = total number of authorised commands issued
Sampling Window	Per validation period and cumulative
Acceptance Threshold	≥ 95% of authorised commands executed successfully
Applicable use cases	UC-1 (relay control), UC-2 (BMS set-point), UC-4 (portfolio)

4.2.5 T-KPI-5: System Availability

Measures the proportion of time that the end-to-end SEMDR system - from edge data collection through model serving to dashboard - is fully operational. Excludes planned maintenance windows agreed in advance with the consortium.

T-KPI-5: System Availability	
Formula	$\boldsymbol{System\ Availability}\ (\%) = \frac{\boldsymbol{T_{operational} - T_{unplanned\ downtime}}}{\boldsymbol{T_{operational}}} \times \mathbf{100}$ $T_{operational}$ = total scheduled operational time excluding planned maintenance $T_{unplanned\ downtime}$ = time lost due to unplanned outages
Sampling Window	Monthly and cumulative project total
Acceptance Threshold	≥ 95% uptime per month; no single outage exceeding 4 hours
Applicable use cases	All (UC-1, UC-2, UC-3, UC-4)

4.3 Energy and Cost KPIs

4.3.1 E-KPI-1: Pool Pump Daily Energy Reduction

Measures the reduction in daily energy consumption of the pool circulation pump attributable to duty-cycle optimisation, compared to the pre-intervention baseline.

E-KPI-1: Pool Pump Daily Energy Reduction	
Formula	$\Delta E_{pump} = E_{baseline,\ pump} - E_{actual,\ pump}\ [kWh/day]$ $E_{baseline_pump}$ = baseline daily pool pump energy consumption E_{actual_pump} = actual daily pool pump energy consumption during the intervention period
Baseline	288 kWh/day
Sampling Window	Per operational day with a completed duty-cycle intervention
Acceptance Threshold	$Mean\ \Delta E_{pump}$≥ 6 kWh/day (equivalent to ≥ 30 minutes reduction per day, 2.1% of baseline) over the validation period
Applicable use cases	UC-1

4.3.2 E-KPI-2: Peak Load Reduction

Measures the reduction in the hotel's peak mains demand during a demand response event, compared to the forecasted demand for the same period in the absence of intervention. Uses model-counterfactual comparison.

E-KPI-2: Peak Load Reduction	
Formula	$\Delta P_{peak} = P_{forecast,\ no\ intervention,e} - P_{actual,e}\ [kW]$ $P_{forecast,\ no\ intervention,e}$ = forecasted peak demand for event $P_{actual,e}$ = actual mains demand during demand response event e e = demand response event
Baseline	Daily peak demand characteristically in the 165-180 kW range (from mains telemetry)
Sampling Window	Per demand response event
Acceptance Threshold	$Mean\ \Delta P_{peak}$≥ 10% of prevailing demand across qualifying demand response events
Applicable use cases	UC-1 (partial), UC-4 (primary)

4.3.3 E-KPI-3: Overall Energy Savings

Measures the total reduction in hotel energy consumption attributable to SEMDR interventions across all active use cases, compared to the weather normalised baseline.

E-KPI-3: Overall Energy Savings	
Formula	$\Delta E_{total} = E_{baseline,\ norm} - E_{actual}\ [kWh]$ $E_{baseline,\ norm}$ = weather normalised baseline energy over the validation period E_{actual} = actual measured energy over the same period
Baseline	Mains consumption profile from August-September 2025, normalised seasonally
Sampling Window	Monthly aggregate, then full validation period (Months 8-11)
Acceptance Threshold	3-5% reduction in total energy consumption of instrumented connected assets over the validation period
Applicable use cases	UC-1, UC-2, UC-4

4.3.4 E-KPI-4: Tariff-Weighted Cost Savings

Measures the reduction in the estimated electricity bill attributable to SEMDR interventions, accounting for time-of-use tariff differentials and peak demand charges.

E-KPI-4: Tariff-Weighted Cost Savings	
Formula	$$\Delta C = \sum_{p=1}^{N} (\Delta E_p \tau_p) + \Delta P_{peak} \tau_{peak} \ (€)$$ ΔE_p = energy reduction in tariff period p (kWh) τ_p = applicable energy tariff in period p (kWh) ΔP_{peak} = reduction in billed peak demand (kW) τ_{peak} = peak demand charge (€/kWh)
Reference Tariff Range	€0.18-0.22/kWh (time-of-use rate applicable to hotel at time of data collection)
Sampling Window	Monthly
Acceptance Threshold	Positive net savings across each month of validated operation
Applicable use cases	UC-1, UC-4

4.3.5 E-KPI-5: Peak-to-Average Ratio Improvement

Measures the change in the ratio of peak mains demand to average mains demand over a rolling 24-hour period. A reduction in this ratio indicates that demand has been flattened, reducing stress on both the hotel's supply connection and the grid.

E-KPI-5: Peak-to-Average Ratio Improvement	
Formula	$$PAR = \frac{P_{max,24h}}{P_{mean,24h}}$$ $$\Delta PAR = PAR_{baseline} - PAR_{intervention}$$ $P_{max,24h}$ = maximum mains demand over the rolling 24-hour window $P_{mean,24h}$ = mean mains demand over the same window
Sampling Window	Daily (rolling 24-hour window), reported weekly
Acceptance Threshold	Mean $\Delta PAR > 0$ (i.e., ratio improves or is maintained) over the validation period; no deterioration in PAR attributable to system operation
Applicable use cases	UC-4

4.4 Service and Comfort KPIs

4.4.1 S-KPI-1: Pool Water Quality Compliance

Measures the proportion of time during which pool water quality parameters remain within the safe operational range during and after duty-cycle interventions.

S-KPI-1: Pool Water Quality Compliance	
Formula	$$\boldsymbol{ORP\ Compliance}\ (\%) = \frac{\boldsymbol{T_{ORP \geq 650mV}}}{\boldsymbol{T_{intervention}}} \times \mathbf{100}$$ $$\boldsymbol{pH\ Compliance}\ (\%) = \frac{\boldsymbol{T_{7.2 \leq pH \leq 7.8}}}{\boldsymbol{T_{intervention}}} \times \mathbf{100}$$ $T_{intervention}$ = total intervention time
Sampling Window	Per intervention session and cumulative
Acceptance Threshold	ORP ≥ 650 mV for ≥ 98% of intervention time; pH within 7.2-7.8 for ≥ 98% of intervention time [1, 6]
Applicable use cases	UC-1

4.4.2 S-KPI-2: Guest Thermal Comfort Compliance

Measures the proportion of service hours during which the lobby temperature remains within the defined comfort band. This is the primary proxy for guest experience quality during HVAC interventions [2].

S-KPI-2: Guest Thermal Comfort Compliance	
Formula	$$\boldsymbol{Thermal\ Comfort\ Compliance}\ (\%) = \frac{\boldsymbol{T_{22°C \leq T_{lobby} \leq 24°C}}}{\boldsymbol{T_{service}}} \times \mathbf{100}$$ $T_{service}$ = total service hour duration
Sampling Window	Daily during any period when UC-2 or UC-4 interventions are active
Acceptance Threshold	≥ 95% of service hours within comfort band across the validation period
Applicable use cases	UC-2, UC-4

4.4.3 S-KPI-3: HVAC Set-Point Deviation Limit

Verifies that the system never exceeds the agreed maximum deviation from the nominal HVAC set-point during any intervention, regardless of energy-saving opportunity.

S-KPI-3: HVAC Set-Point Deviation Limit	
Formula	$$\boldsymbol{\Delta T_{setpoint,max} = \max_{t \in \mathcal{J}} \left\| T_{setpoint,applied}(t) - T_{setpoint,nominal}(t) \right\|}$$ $\mathcal{J}$ = all intervention periods
Sampling Window	Per intervention; cumulative maximum
Acceptance Threshold	Maximum deviation ≤ 1°C at all times. Any breach constitutes an automatic intervention failure and triggers a protocol review.
Applicable use cases	UC-2

4.4.4 S-KPI-4: Refrigeration Temperature Safety (Monitoring)

Continuous refrigeration temperature monitoring is maintained as a safety and anomaly-detection measure. Any temperature deviation outside safe food-storage bands triggers an immediate operational alert.

S-KPI-4: Refridgeration Temperature Safety	
Formula	$$\boldsymbol{Refrigeration\ Complience}\ (\%) = \frac{T_{T_{fridge} \in \mathcal{H}}}{T_{monitoring}} \times 100$$ $\mathcal{H}$ = HACCP safe temperature band $T_{monitoring}$ = total monitoring time
Sampling Window	Continuous; daily report
Acceptance Threshold	100% compliance with HACCP temperature bounds [1] at all times. Any exceedance must be flagged, investigated, and resolved within 2 hours.
Applicable use cases	UC-3

4.4.5 S-KPI-5: Operational Staff Workload (All use cases)

Assesses whether the SEMDR system creates additional operational burden for hotel staff compared to baseline. This KPI is assessed qualitatively through a structured periodic survey of hotel operations personnel.

S-KPI-5: Operational Staff Workload	
Formula	$$\overline{W} = \frac{1}{N}\sum_{i=1}^{N} W_i$$ $$\overline{U} = \frac{1}{N}\sum_{i=1}^{N} U_i$$ W_i = workload impact score from respondent i on the 1-5 Likert scale U_i = usability score from respondent i N = number of completed questionnaires in the sampling window
Sampling Window	Monthly throughout the validation period
Acceptance Threshold	Mean staff workload impact score ≤ 2.5/5 (neutral to positive); system usability score ≥ 3.5/5. Negative trends trigger a user experience review.
Applicable use cases	All

4.5 KPI Summary Table

KPI	Formula / Metric	Sampling Window	Acceptance Threshold	Applicable Use Cases
T-KPI-1: Data Availability	$\frac{N_{valid,\ received}}{N_{expected}} \times 100$	Rolling 24h & 7d	≥ 97% per rolling 7d	All
T-KPI-2: E2C Latency	$P_{95}(t_{cloud_arrival} - t_{edge_acquisition})$	Rolling 1h	95th pctile ≤ 10s	UC-1, 2, 4
T-KPI-3: Forecast MAPE	$\frac{1}{N}\sum_{i=1}^{N}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert \times 100$	Rolling 7d	MAPE ≤ 8%	UC-4, UC-1
T-KPI-4: Actuation Rate	$\frac{N_{successful,\ confirmed}}{N_{authorised,\ issued}} \times 100$	Per period & cumulative	≥ 95%	UC-1, 2, 4
T-KPI-5: System Uptime	$\frac{T_{operational} - T_{unplanned\ downtime}}{T_{operational}} \times 100$	Monthly	≥ 95%/month, <4h/outage	All
E-KPI-1: Pump Daily ΔE	$E_{baseline,\ pump} - E_{actual,\ pump}\ [kWh/day]$	Per intervention day	Mean ≥ 6 kWh/day	UC-1
E-KPI-2: Peak Load Reduction	$P_{forecast,\ no\ intervention,e} - P_{actual,e}\ [kW]$	Per DR event	Mean ≥ 10% of prevailing load	UC-1, UC-4
E-KPI-3: Total Energy Saving	$E_{baseline,\ norm} - E_{actual}\ [kWh]$	Monthly + full period	3-5% of connected assets	UC-1, 2, 4
E-KPI-4: Cost Savings	$\sum_{p=1}^{N}(\Delta E_p \tau_p) + \Delta P_{peak}\tau_{peak}\ (€)$	Monthly	Net positive; payback < 24 months	UC-1, UC-4
E-KPI-5: PAR Improvement	$PAR_{baseline} - PAR_{intervention}$	Daily (rolling 24h)	Mean ΔPAR > 0	UC-4
S-KPI-1: Pool Water Quality	$\frac{ORP\ \geq\ 650\ mV\ AND\ pH\ 7.2 - 7.8}{Total} \times 100$	Per session & cumulative	≥ 98% compliance	UC-1
S-KPI-2: Lobby Comfort	$\frac{T_{22°C \leq T_{lobby} \leq 24°C}}{T_{service}} \times 100$	Daily during UC-2/4	≥ 95% of service hours	UC-2, UC-4
S-KPI-3: Set-Point Deviation	$\max_{t \in \mathcal{J}}\lvert T_{setpoint,applied}(t) - T_{setpoint,nominal}(t)\rvert$	Per intervention	≤ 1°C at all times	UC-2
S-KPI-4: Fridge Safety	$\frac{T_{T_{fridge} \in \mathcal{H}}}{T_{monitoring}} \times 100$	Continuous / daily	100% compliance (monitoring only)	UC-3 (monitoring)
S-KPI-5: Staff Workload	**Likert survey score (1-5)**	Monthly	Workload ≤ 2.5; Usability ≥ 3.5	All

Table 4 - KPI Summary Table

5 Testing and Validation Methodology

5.1 Staged Validation Protocol

The SEMDR validation follows a gated, four-stage progression. This approach was adopted to protect hotel operations and guest comfort at all times while building the empirical evidence base needed for autonomous control. Progression from one stage to the next requires explicit confirmation that the promotion criteria have been met, reviewed jointly by HO and PA1.

Stage	Description	Control Mode	Promotion Criteria
Stage 0: Baseline Collection	Continuous telemetry from all instrumented assets. No interventions. Establish stable baseline distributions for all KPIs.	Read-only monitoring. No actuation.	Minimum 4 weeks of complete, quality-checked data for each asset. T-KPI-1 ≥ 97% for all streams.
Stage 1: Shadow Mode	Models generate recommendations and log what they would do, without any actuation. Recommendations visible to operators on dashboard.	Models active; no physical actuation. Operator review only.	T-KPI-3 MAPE ≤ 8% on rolling 7-day window. T-KPI-5 ≥ 95% system uptime. No open data quality alerts.
Stage 2: Supervised Actuation	Selected interventions executed under direct human oversight, with hotel staff present or on standby. Operator confirms each action via dashboard.	Model proposes; operator authorises each individual action.	T-KPI-4 ≥ 95% actuation success over first 10 supervised events. All safeguard limits respected. No guest-visible incidents.
Stage 3: Controlled Autonomous	System executes pre-approved intervention types autonomously within defined guardrails. Operator can suspend at any time.	Autonomous within guardrails; operator retains override at all times.	Sustained performance across ≥ 4 weeks in Stage 2. All KPIs trending positive. Formal sign-off by HO and hotel operations management.

Table 5 - Staged Validation Protocol

At the time of this document (M6, September 2025), the current status of each use case against this protocol is as follows:

- UC-1 (pool pump) has completed Stage 0 (baseline collection from August 2025, M5) and has executed Stage 2 supervised duty-cycle trials (September 2025, M6). Shadow-mode demand forecast and duty-cycle models are running on live data. Progression to Stage 3 (controlled autonomous operation) is anticipated once the algorithm has been validated in extended shadow mode and the promotion criteria are confirmed, targeted for the M8-M9 period (November-December 2025).
- UC-2 (HVAC/chiller) has BMS upgrade completed M6; metering installation in progress; power telemetry expected upon completion. Stage 0 baseline collection for chiller power consumption will commence once rooftop feed metering is operational, expected from M7 (October 2025). Shadow-mode operation is targeted for M8-M9 (November-December 2025).

- UC-3 (refrigeration) has been in continuous monitoring since M5 (August 2025). Assessment of DR viability is ongoing. If refrigeration is confirmed as a viable DR target, it will enter the staged validation protocol; otherwise, monitoring will continue for anomaly detection purposes.
- UC-4 (demand shaping) will enter Stage 1 once both UC-1 and UC-2 have progressed sufficiently through their respective validation stages to provide reliable individual control levers.

5.2 Baseline Construction

Accurate baselines are essential for a credible before/after comparison. The following methodology will be applied:

- For mains demand and pool pump energy: rolling 28-day baseline calculated from actual telemetry in the period immediately preceding the validation window (August-September 2025 data for early validation, updated monthly). Pool pump baseline confirmed at approximately 288 kWh/day from August 2025 (M5) telemetry.
- For chiller consumption: baseline will be established from the first 4 full weeks of production-quality chiller power telemetry (expected once metering installation is complete), explicitly noted in D3.4. BMS supervisory data will provide supplementary operational context once the upgrade is fully commissioned.
- Normalisation: where weather-driven demand variation is material (particularly for HVAC), daily cooling degree-day normalisation will be applied using a base temperature of 22°C calculated from AccuWeather actuals. Occupancy index (rooms occupied / total rooms), provided weekly by hotel management under the data sharing protocol established in M2, will be included as a normalisation covariate. In the event that occupancy data is unavailable for specific periods, lobby temperature and humidity patterns will serve as a proxy indicator of occupancy-driven demand.
- Counterfactual baseline for event-level KPIs (E-KPI-2, E-KPI-5): constructed from the short-horizon demand forecast for the same period, representing what demand would have been without the intervention. The forecast model's accuracy is bounded by T-KPI-3.

5.3 Measurement and Data Quality Requirements

All KPI measurements must meet the following data quality standards to be included in the D3.4 assessment:

- Data completeness: T-KPI-1 ≥ 97% for the measurement period. Days or periods with completeness below 90% are excluded from KPI calculations and noted in D3.4.
- Timestamp integrity: all records must carry a synchronised UTC timestamp from the edge device time server. Time drift greater than 5 seconds triggers an alert and the affected window is quarantined until resynchronisation is confirmed.

- Outlier handling: values flagged by the automated outlier detection at ingestion are excluded from KPI calculations. A sustained anomaly on any stream (more than 3 consecutive flagged samples) triggers an engineering review before the stream is restored to the production KPI dataset.
- Intervention logging: every actuation event must be logged with: model version, feature input hash, recommended action, operator authorisation (if applicable), command issued, and telemetry-confirmed outcome. Unlogged events are excluded from the actuation success rate KPI.
- Minimum intervention sample for E-KPIs: at least 15 qualifying intervention events per use case during the validation period (Months 8-11, November 2025 - February 2026) are required for statistical significance. Where fewer events are available for a use case due to operational constraints or compressed validation windows (particularly for UC-2 and UC-4, which depend on chiller telemetry availability once metering installation is complete) this will be clearly noted in D3.4 with appropriate confidence qualifications.

5.4 Acceptance Criteria and Pass/Fail Logic

A use case is considered to have passed validation if all of the following conditions are met:

1. All Technical KPIs (T-KPI-1 through T-KPI-5) relevant to the use case meet their thresholds throughout the validation period.
2. At least one Energy KPI (E-KPI-1 through E-KPI-5) directly associated with the use case meets its acceptance threshold.
3. All Service and Comfort KPIs (S-KPI-1 through S-KPI-5) relevant to the use case meet their thresholds throughout the validation period. A single breach of a hard safety KPI (S-KPI-3, S-KPI-4) constitutes an automatic failure for the associated use case regardless of energy results.

A use case is considered to have partially passed if technical and safety KPIs are met but the energy KPI target falls below threshold. In this case D3.4 will include a root-cause analysis, a projection of what energy savings would be achievable under extended operation or adjusted conditions, and a recommendation on whether the shortfall is a system design issue or an operating conditions issue.

The overall SEMDR system validation passes if UC-1 and at least one of UC-2 or UC-4 achieve full or partial pass status, given that UC-1 is the most mature and best-evidenced use case and that chiller data availability introduces a timing dependency for UC-2 and UC-4.

6 Conclusion

This document defines the complete use case scenario and KPI framework for the SEMDR project validation at pilot site. Four use cases have been specified: pool-pump duty-cycle optimisation (UC-1), HVAC supervisory set-point tuning (UC-2), refrigeration efficiency investigation (UC-3), and peak-period demand shaping (UC-4).

The KPI framework establishes 15 specific indicators across three layers: 5 technical KPIs confirming platform health, 5 energy and cost KPIs quantifying demand response impact, and 5 service and comfort KPIs protecting guest experience. Every KPI includes a formula, sampling window, and acceptance threshold, enabling a rigorous and unambiguous like-for-like assessment in D3.4.

The baseline measurements recorded in Section 3.3 - including the pool pump's 288 kWh/day consumption baseline (860 m^3 pool, 12 kW, 80 m^3/h), the mains 135-180 kW daily band, and the confirmed pool water quality stability during M6 supervised trials - provide a credible and transparent reference for all subsequent comparisons. Conservative hand-calculated savings estimates indicate a range of €32-160 per month from pool pump optimisation alone, with additional savings expected from tariff-aware scheduling and HVAC interventions once chiller metering is operational.

The staged validation protocol (Section 6.1) defines gated progression from baseline collection through shadow operation, supervised actuation, and controlled autonomous operation. UC-1 is the most advanced, having completed supervised trials at M6. UC-2 and UC-4 will follow once chiller metering installation is complete. UC-3 remains under active investigation. Deliverable D3.4 (System Validation and Performance Report, due Month 12, March 2026) will use this document as its primary reference, with every KPI result directly traceable to the definitions, formulas, and thresholds established here.

KPI Checkpoint 1 Status (M3.2) - September 2025
KPI Checkpoint 1 (Milestone M3.2) is completed in conjunction with this deliverable at Month 6 (September 2025). The checkpoint confirms: (1) all KPI definitions finalised and agreed by consortium; (2) Stage 0 baseline data collected for mains, pool pump, refrigeration, and lobby comfort - all live since August 2025 (M5); (3) UC-1 supervised duty-cycle trials completed in September 2025 with all safeguard KPIs met across 30-minute to 4-hour off-period durations; (4) chiller BMS upgrade secured and metering installation in progress; (5) forecast models running in shadow mode on live data; (6) chiller BMS upgrade complete and rooftop feed metering installation in progress.

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Appendix A: KPI-to-Proposal Objective Traceability

The following table maps each D3.3 KPI to the specific scientific (SO) and technological (TO) objectives from the SEMDR proposal (COM-CONCEPT-ENERGY/0624), ensuring that the validation framework provides complete coverage of the project's stated commitments.

Proposal Objective	Objective Description	Mapped KPIs	Rationale
SO1	Implement real-time energy monitoring across key hotel systems by M2	T-KPI-1, T-KPI-2, T-KPI-5	SO1 requires workable uptime with secure real-time transmission. These technical KPIs directly measure the monitoring infrastructure's reliability, latency, and availability.
SO2	Enhance demand response algorithms for the hospitality sector by M6	E-KPI-1, E-KPI-2, T-KPI-3, T-KPI-4 S-KPI-2, S-KPI-3	SO2 targets 10% peak reduction while maintaining comfort within ±1.5°C. E-KPI-2 measures peak reduction; S-KPI-2 and S-KPI-3 enforce the comfort constraint. T-KPI-3 and T-KPI-4 confirm the algorithms function as designed.
SO3	Develop and deploy a user-friendly energy management interface by M3	T-KPI-5 S-KPI-5	SO3 requires dashboard completion with staff feedback and manual override under 30 seconds. T-KPI-5 measures system availability including the dashboard. S-KPI-5 captures usability and workload impact through structured staff surveys.
TO1	Quantify energy and cost savings by M12	E-KPI-3, E-KPI-4	TO1 requires 3-5% energy reduction and ROI with payback under 24 months. E-KPI-3 directly measures the energy reduction target. E-KPI-4 quantifies cost savings and provides the basis for the payback calculation in D3.4.
TO2	Optimise system performance for scalability by M9	T-KPI-1, T-KPI-2, T-KPI-4, T-KPI-5	TO2 requires the system to handle high-granularity data without degradation. The technical KPIs collectively verify that the platform performs reliably across multiple concurrent data streams and control actions.
TO3	Validate system reliability and guest satisfaction by M12	T-KPI-5 S-KPI-1, S-KPI-2, S-KPI-4, S-KPI-5	TO3 requires >95% uptime and comfort resolution within 24 hours. T-KPI-5 measures the uptime target directly. The service KPIs confirm that guest comfort, water quality, and food safety are maintained. S-KPI-5 provides the monthly staff satisfaction feedback channel.
TO4	Prepare for future energy market participation by M9	E-KPI-2, E-KPI-4, E-KPI-5	TO4 requires a simulation model exploring flexibility market participation. The energy KPIs provide the empirical inputs for the market simulation: actual demand reduction capability (E-KPI-2), realised cost savings (E-KPI-4), and load-flattening performance (E-KPI-5).